

Assignment. 6

Title: Implement a program to handle Arithmetic exception, Array Index Out Of Bounds. The user enters two numbers Num1 and Num2. The division of Num1 and Num2 is displayed. If Num1 and Num2 were not integers, the program would throw a Number Format Exception. If Num2 were zero, the program would throw an Arithmetic Exception. Display the exception.

Aim: Exception handling

Objective: To apply the concept of exception handling in Java

Theory

Exception:

An Exception is an unwanted event that interrupts the normal flow of the program. When an exception occurs program execution gets terminated. In such cases we get a system generated error message. The good thing about exceptions is that they can be handled in Java. By handling the exceptions we can provide a meaningful message to the user about the issue rather than a system generated message, which may not be understandable to a user.

Major reasons why an exception Occurs

- Invalid user input
- Device failure
- Loss of network connection
- Physical limitations (out-of-disk memory)
- Code errors
- Opening an unavailable file

Exception Handling

Exception handling is one of the most important feature of java programming that allows us to handle the runtime errors caused by exceptions.

Advantage of exception handling

Exception handling ensures that the flow of the program doesn't break when an exception occurs. For example, if a program has bunch of statements and an exception occurs mid way after executing certain statements then the statements after the exception will not execute and the program will terminate abruptly.

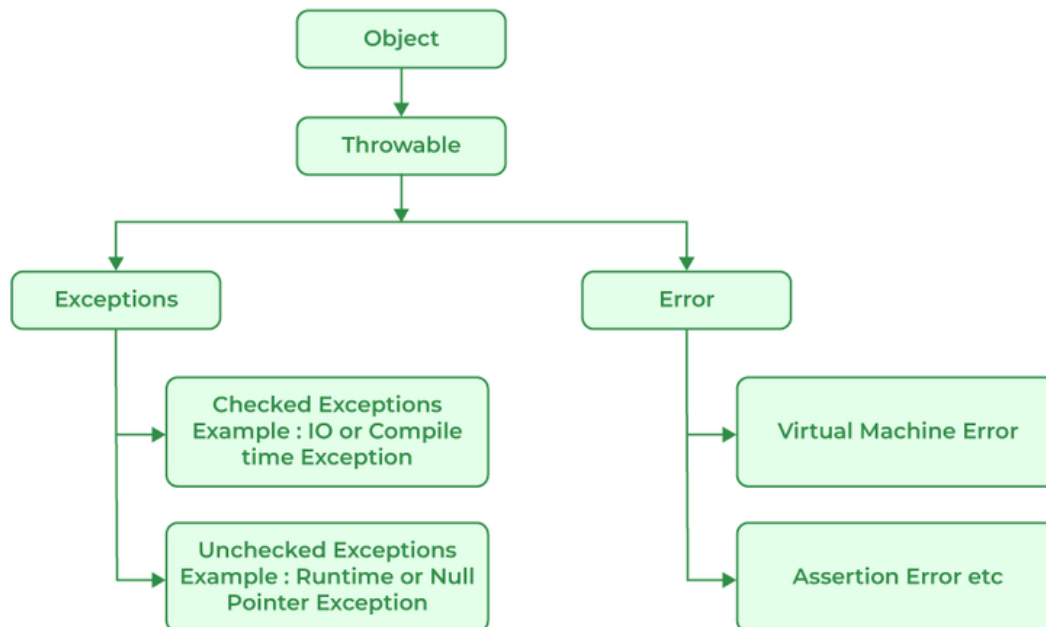
By handling we make sure that all the statements execute and the flow of program doesn't break. Difference between error and exception.Errors indicate that something severe enough has gone wrong, the application should crash rather than try to handle the error.

Exceptions are events that occurs in the code. A programmer can handle such conditions and take necessary corrective actions.

Exception Hierarchy

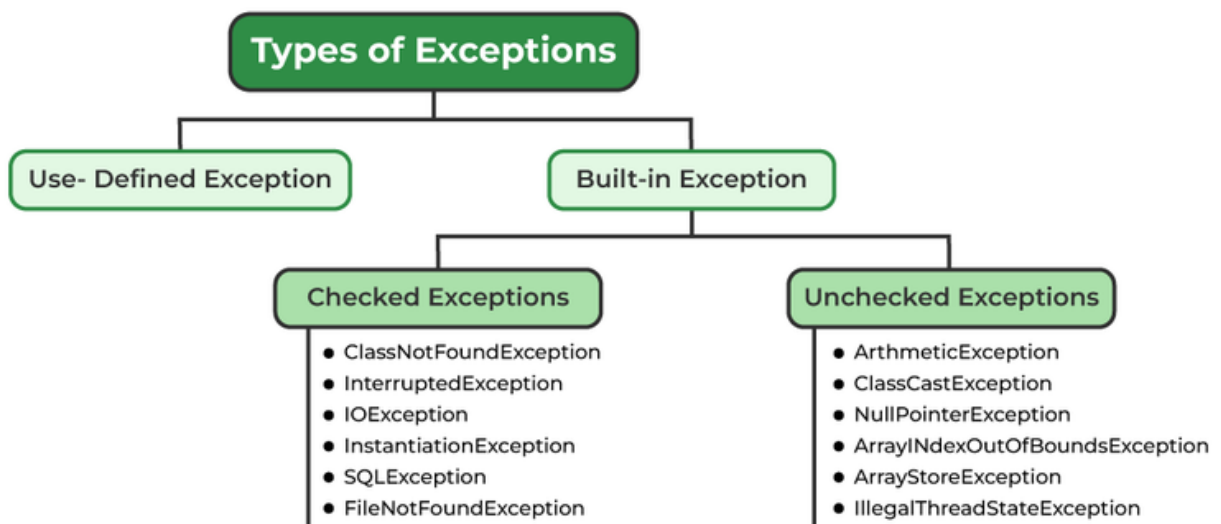
All exception and error types are subclasses of the class Throwable, which is the base class of the hierarchy. One branch is headed by Exception. This class is used for exceptional conditions that user programs should catch.

NullPointerException is an example of such an exception. Another branch, Error is used by the Java run-time system(JVM) to indicate errors having to do with the run-time environment itself(JRE). StackOverflowError is an example of such an error.



Types of Exceptions

Java defines several types of exceptions that relate to its various class libraries. Java also allows users to define their own exceptions.



Checked exceptions

All exceptions other than Runtime Exceptions are known as Checked exceptions as the compiler checks them during compilation to see whether the programmer has handled them or not. If these exceptions are not handled/declared in the program, you will get compilation error. For example, SQLException, IOException, ClassNotFoundException etc.

Unchecked Exceptions

Runtime Exceptions are also known as Unchecked Exceptions. These exceptions are not checked at compile-time so compiler does not check whether the programmer has handled them or not but it's the responsibility of the programmer to handle these exceptions and provide a safe exit.

For example, `ArithmeticException`, `NullPointerException`, `ArrayIndexOutOfBoundsException` etc.

a) **Arithmetic Exception:**

The following are two situations where the arithmetic exception may occur.

- When we perform a division where 0 is used as a divisor, i.e., 0 comes as the denominator.
- A long decimal number that is non-terminating by Big Decimal.

b) **ArrayIndexOutOfBoundsException:**

The `ArrayIndexOutOfBoundsException` is one of the most common errors in Java. It occurs when a program attempts to access an invalid index in an array i.e. an index that is less than 0, or equal to or greater than the length of the array.

Since a Java array has a range of $[0, \text{array length} - 1]$, when an attempt is made to access an index outside this range, an `ArrayIndexOutOfBoundsException` is thrown.

Java Exception Keywords

Java provides five keywords that are used to handle the exception. The following table describes each.

try	The "try" keyword is used to specify a block where we should place an exception code. It means we can't use try block alone. The try block must be followed by either catch or finally.
catch	The "catch" block is used to handle the exception. It must be preceded by try block which means we can't use catch block alone. It can be followed by finally block later.
finally	The "finally" block is used to execute the necessary code of the program. It is executed whether an exception is handled or not.
throw	The "throw" keyword is used to throw an exception.
throws	The "throws" keyword is used to declare exceptions. It specifies that there may occur an exception in the method. It doesn't throw an exception. It is always used with method signature.

Input: Enter the number Num1 & Num2

Output: Displays the exception

Algorithm:

Step1: Start.

Step 2: Create a try-catch block to handle exceptions.

Inside the try block: Prompt the user to enter two numbers, Num1 and Num2.

Step 3: Read Num1 and Num2 from the user.

Step 4: Check for the following conditions:

- i. If Num2 is equal to 0, throw an Arithmetic Exception with a message indicating that division by zero is not allowed.
- ii. If Num1 or Num2 are not integers (e.g., the user enters non-integer values), throw a Number Format Exception with a message indicating that valid integers should be entered.
- iii. If no exceptions are thrown, perform the division operation:

Step 5: In the catch block, catch any ArithmeticException, Array Index Out of Bounds Exception and NumberFormatException, and display the appropriate exception message to the user.

Step 6: Stop

Conclusion

Thus we have studied exception handling concept using java

Questions:

1. What is the main purpose of the program you've implemented, and what kind of exceptions does it handle?
2. How does the program handle the situation when the user enters non-integer values for Num1 and Num2?
3. What happens if the user enters a value of 0 for Num2? How is this situation handled by the program?
4. Can you explain how the program uses a try-catch block to handle exceptions, and what messages are displayed to the user in case of exceptions?
5. Does the program handle Array Index Out of Bounds exceptions, and if so, in what context is this exception handled?

(write answer for the above questions) & Attached code with output